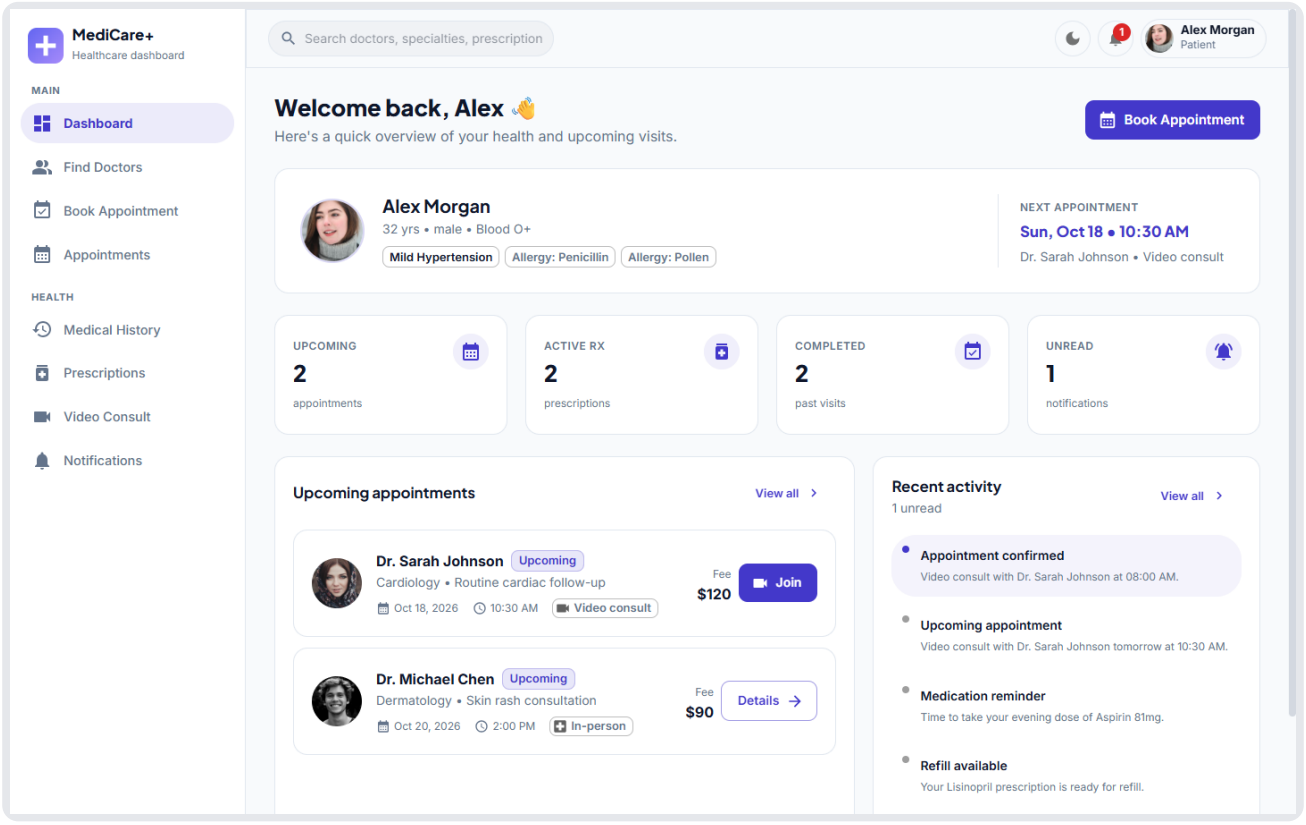
MediCare+

PROJECT REPORT

Doctor Appointment Booking System

What it is, how to use it, how it works, and how to run it

Author: Sabin Ghimire | Version 1.0 | 17 October 2026



The patient dashboard

Source code: github.com/Leosabinghimire/Doctor-Appointment-Booking-System

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1. Summary

MediCare+ is a complete web application for a medical clinic. **Patients** use it to find a doctor, see real open times, and book a visit online. **Clinic staff (admins)** use it to manage doctors, their working hours and time off, see every booking, and follow phone calls handled by an AI voice assistant.

The system has three parts: a website that runs in the browser (the **frontend**), a server that holds all the rules and checks who is allowed to do what (the **backend**), and a **database** where everything is saved. An optional connection to **Telentir**, an AI phone-agent service, can call patients with appointment reminders.

Area	What it is
Website (frontend)	React 18 + TypeScript + Material UI. Works on desktop and mobile, light and dark mode.
Server (backend)	Node.js + TypeScript + Express. A REST API that enforces every rule.
Database	SQLite — a single file (medicare.db). Nothing extra to install.
Accounts	Patients sign up themselves; admins are created by the clinic. Passwords are stored securely.
AI phone agent	Optional Telentir integration: reminder calls, call history, transcripts.
Source code	github.com/Leosabinghimire/Doctor-Appointment-Booking-System

> **Project status**
> Everything described in this report has been built and tested in a browser, except the live Telentir phone calls, which need a Telentir account to test. Parts of the screen that are not finished yet are listed honestly in section 12 “Current limitations”.

2. Features

For patients

- **Create an account and sign in** — with email and password.
- **Find doctors** — search by name, specialty or hospital; filter by city, fee, video visits and “available today”.
- **Doctor profiles** — qualifications, experience, hospital, languages, fees and the doctor’s real next open time.
- **Book an appointment** in four simple steps. A time slot can never be booked twice.
- **My appointments** — upcoming, completed and cancelled visits; cancel an upcoming visit.
- **Dashboard** — next appointment, visit counts, active prescriptions and recent notifications.
- **Medical history and prescriptions** — the patient’s own records (read-only).
- **Notifications** — for example “Appointment confirmed” right after booking.

For clinic admins

- **Admin dashboard** — live totals: patients, doctors, today’s bookings, revenue this month; charts of bookings per day and per specialty; newest sign-ups.
- **All appointments** — every booking in the clinic; book on behalf of a patient.
- **Manage doctors** — add, edit, archive and restore doctors; upload a photo.
- **Schedules** — each doctor’s weekly working hours (with a break) and time off such as holidays.
- **Call logs** — phone calls handled by the Telentir AI agent, with full written transcripts.
- **Reminder calls** — one click asks the AI agent to phone a patient about their appointment.

Everywhere

- Clean, modern design with light and dark mode, and a layout that adapts to phones.
- Clear messages for every mistake (for example “That time slot was just taken”).
- Every permission is checked by the server, not only hidden in the screen.

3. User guide — patients

This section walks through the website as a patient sees it. All screenshots are from the running application.

3.1 Sign in or create an account

Every page needs signing in. New patients click **Create account** and enter their name, email, phone number and a password of at least 8 characters. After 5 wrong passwords, sign-in for that email from that network is paused for 15 minutes to stop guessing.

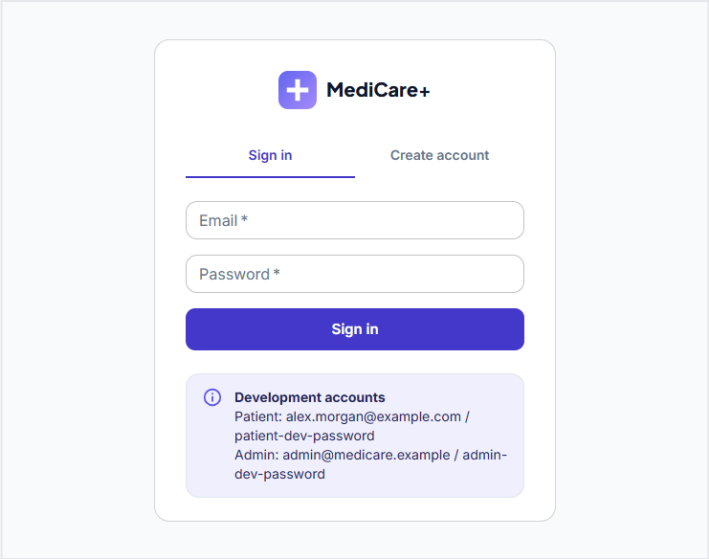


Figure 1 — Sign-in page (the blue box with test accounts is shown only in development)

3.2 Dashboard

After signing in, the patient sees their name, health details (allergies, conditions), their **next appointment**, counts of upcoming and completed visits, active prescriptions and the latest notifications.

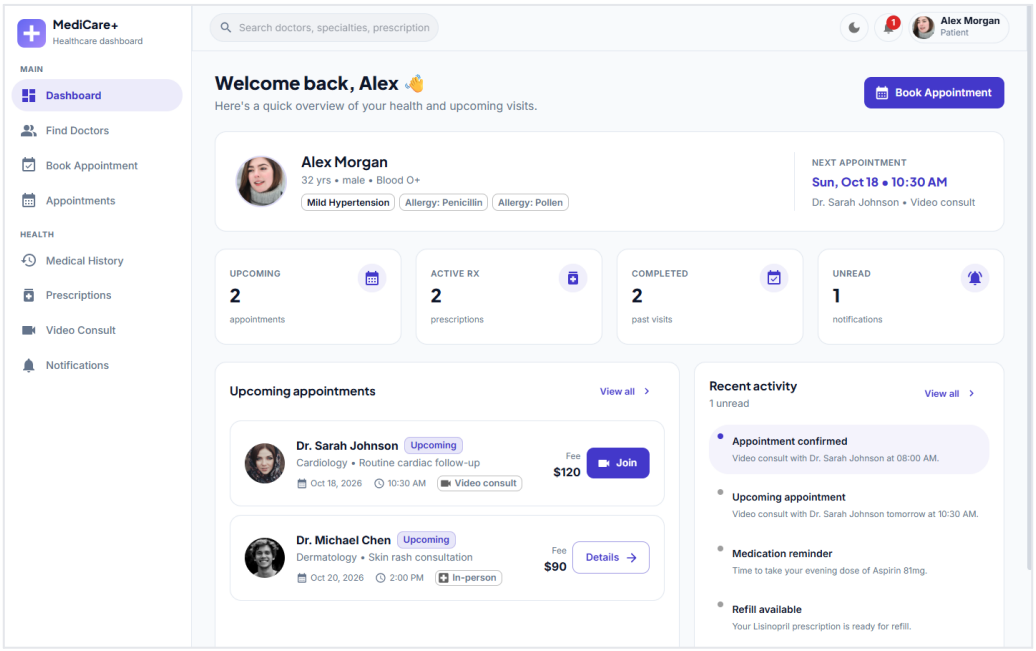


Figure 2 — Patient dashboard

3.3 Find a doctor

Find Doctors lists every doctor the clinic has published. The patient can search and filter. Each card shows the doctor's next open time, which is calculated from the real schedule.

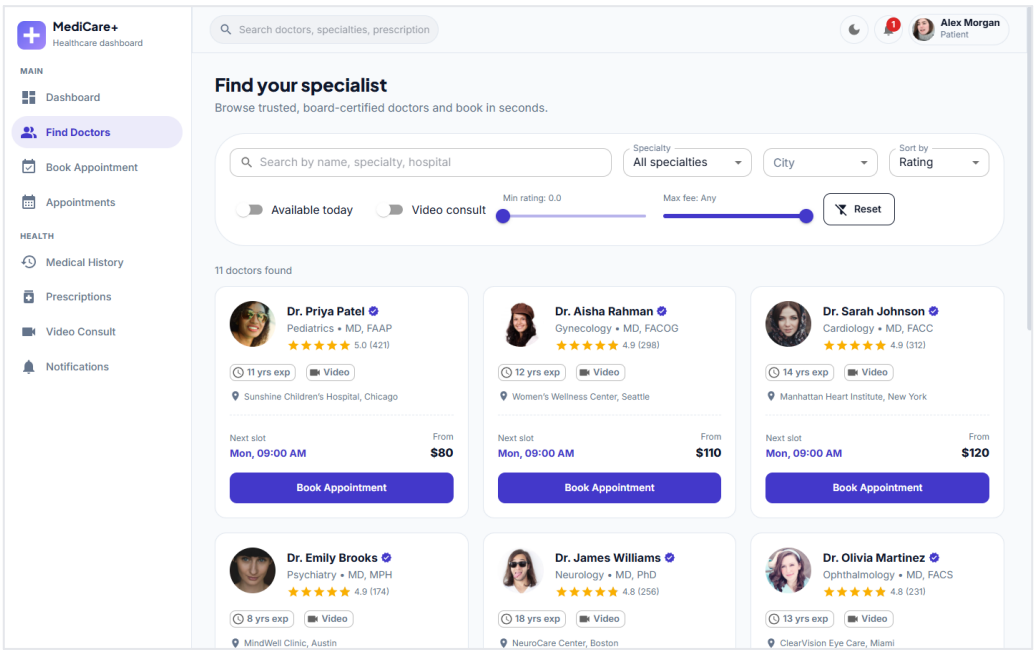


Figure 3 — Find Doctors with search and filters

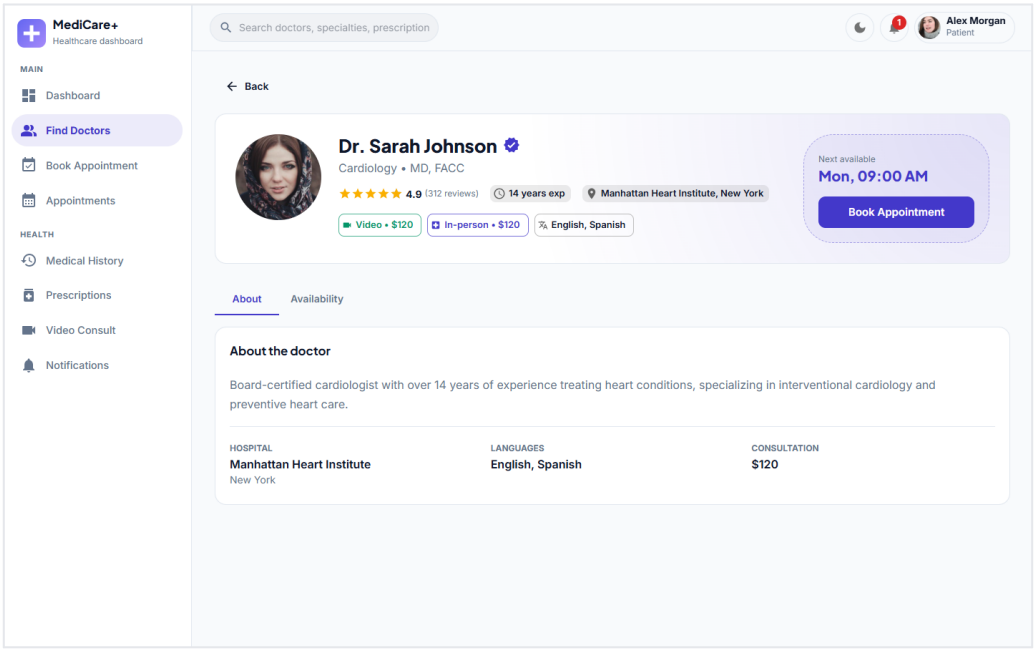
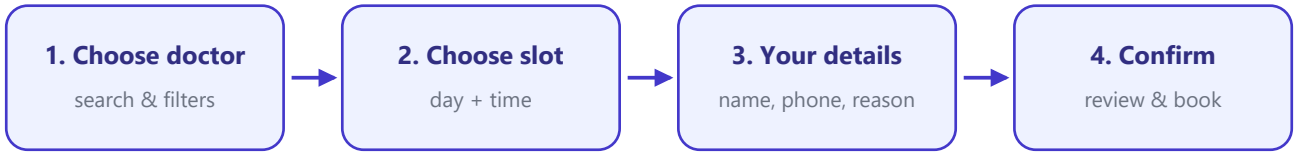


Figure 4 — Doctor profile

3.4 Book an appointment



1. Choose a doctor (or click **Book Appointment** on a doctor's card).
2. Choose **Video consult** or **In-person**, then a day and a time. Each day shows how many times are free. **Closed** means the doctor doesn't work that day; **Full** means every time is taken.
3. Check your name, phone and email (filled in automatically) and write the reason for the visit.
4. Review and click **Confirm Booking**. A confirmation appears in **Notifications**.

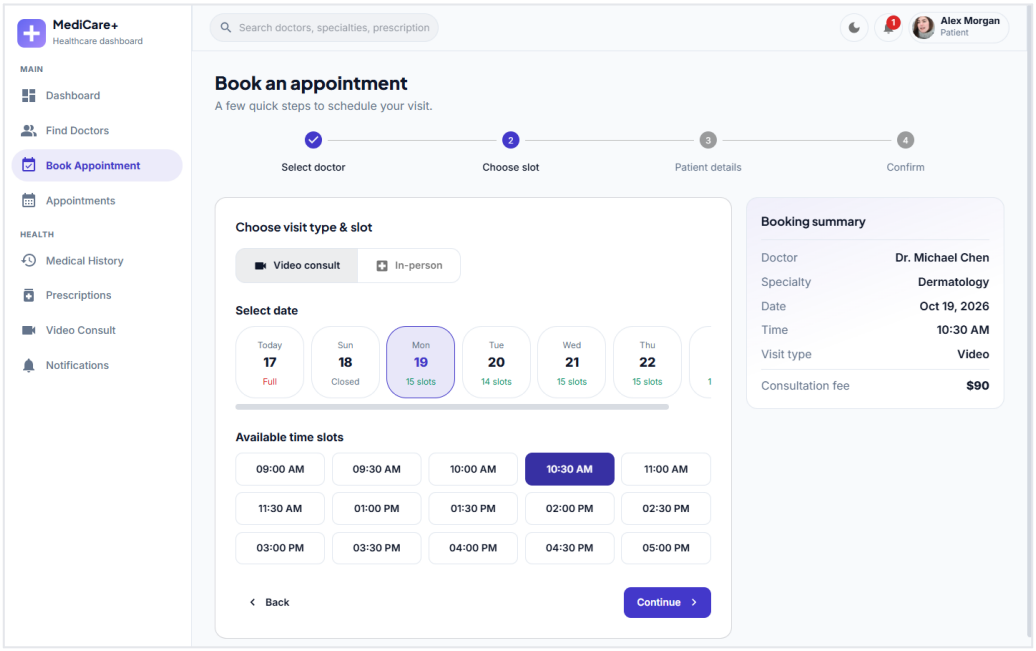


Figure 5 — Choosing a day and time. The summary on the right updates as you choose.

3.5 My appointments

Shows upcoming, completed and cancelled visits. An upcoming visit can be cancelled with the red button; the time slot then becomes free for other patients.

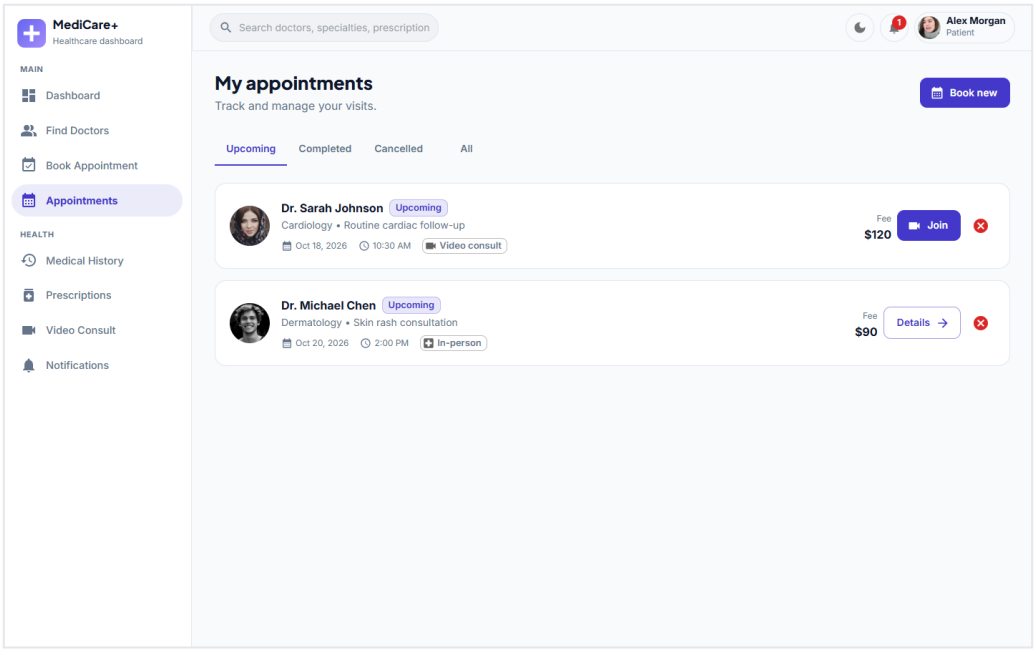


Figure 6 — My appointments

3.6 Prescriptions and medical history

Patients can read their prescriptions (medicines, dose, how often, refills left) and their medical history. These records are entered by the clinic; patients cannot change them.

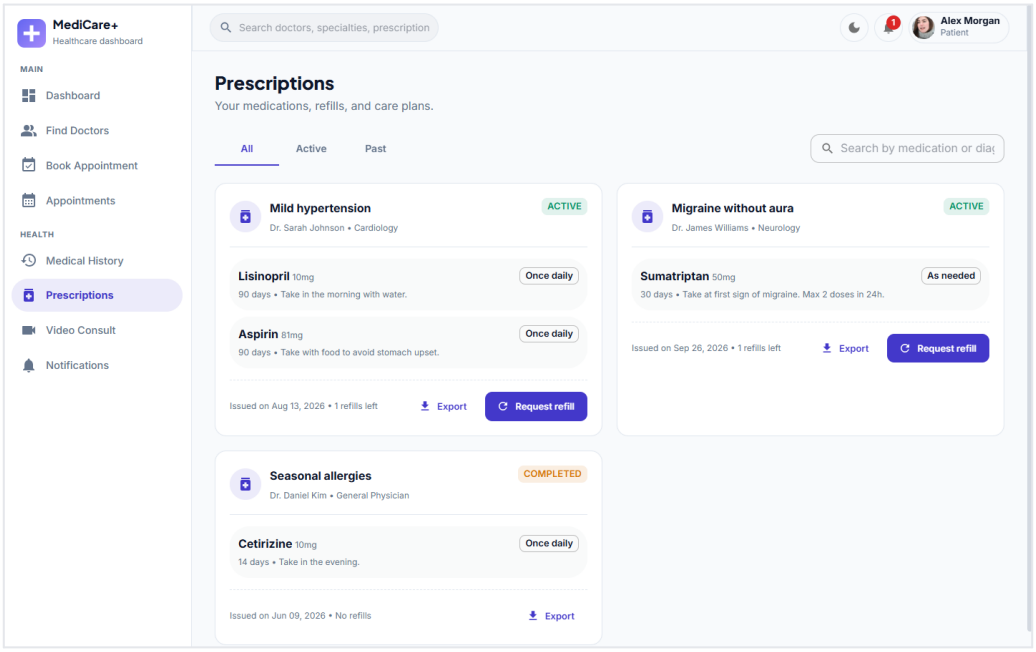


Figure 7 — Prescriptions

4. User guide — clinic admins

Admins sign in on the same page. They see a different menu: Find Doctors, Book Appointment, Appointments, Admin Panel, Manage Doctors and Call Logs.

4.1 Admin dashboard

All numbers are calculated live from the database: total patients and doctors, bookings today, revenue this month, bookings per day over the last week, bookings per specialty, and the newest patient sign-ups.

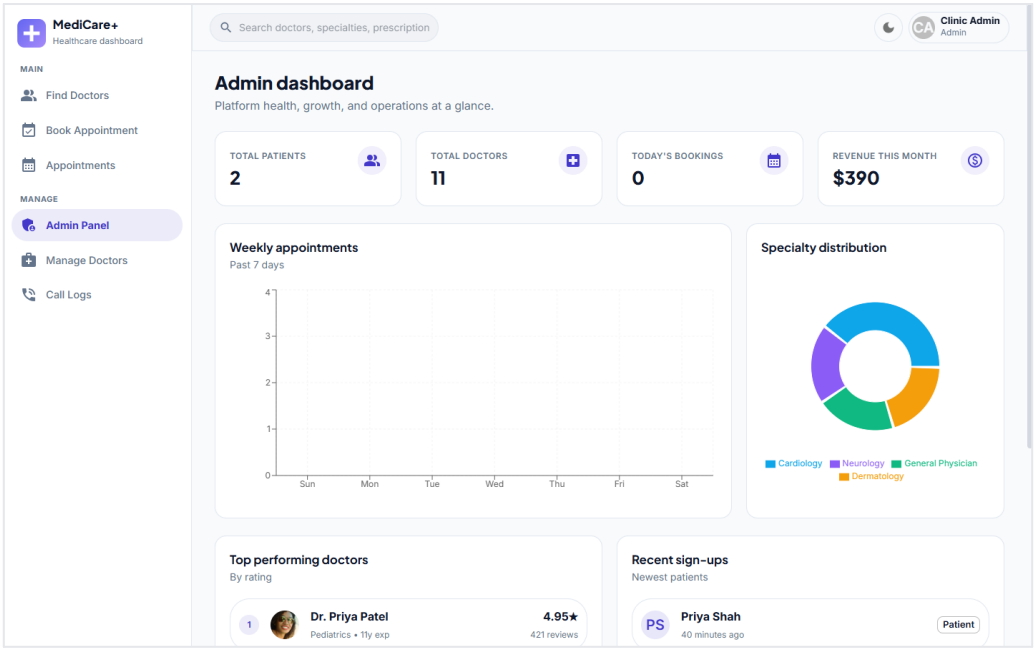


Figure 8 — Admin dashboard

4.2 All appointments

Admins see every booking in the clinic. The phone icon on an upcoming visit asks the AI agent to call the patient with a reminder (needs Telentir).

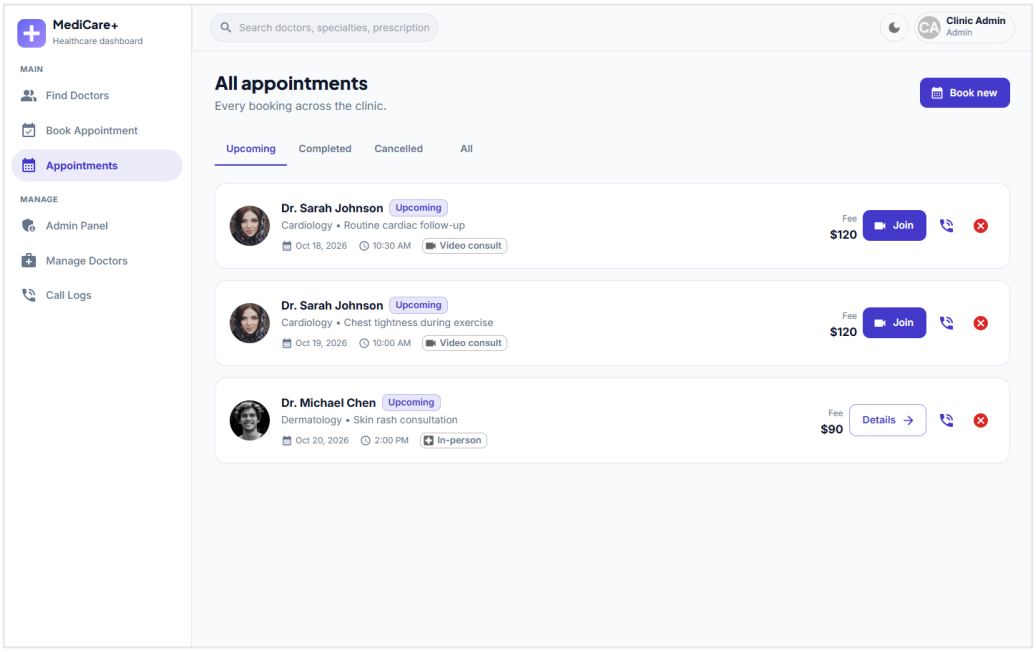


Figure 9 — All appointments (admin view)

4.3 Manage doctors

The list shows every doctor with fee, visit types and how many upcoming visits they have. The icons on each row open the **schedule**, **edit** the doctor, or **archive** them.

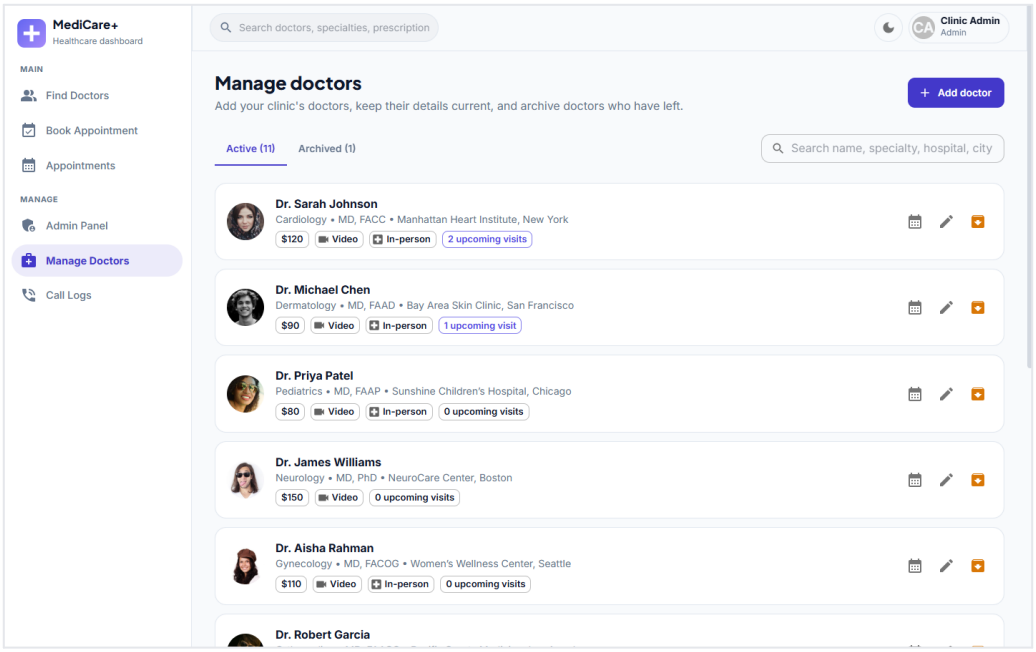


Figure 10 — Manage Doctors

- **Add or edit a doctor** — name, specialty, qualifications, experience, fee, hospital, city, languages, a short “about” text, and video and/or in-person visits. If something is missing, every problem is listed at once.

- **Photo** — click **Upload photo** and choose a JPEG, PNG or WebP image. It is shrunk automatically (a large phone photo becomes about 50 KB) and saved inside the database. A web link to a photo also works.
- **Archive** — for a doctor who has left. They disappear from patient search and booking, but their past and already-booked visits are kept. The warning tells the admin how many upcoming visits the doctor still has. Archived doctors can be restored.
- **New doctors** show the label “New” instead of a star rating, because there is no review system yet.

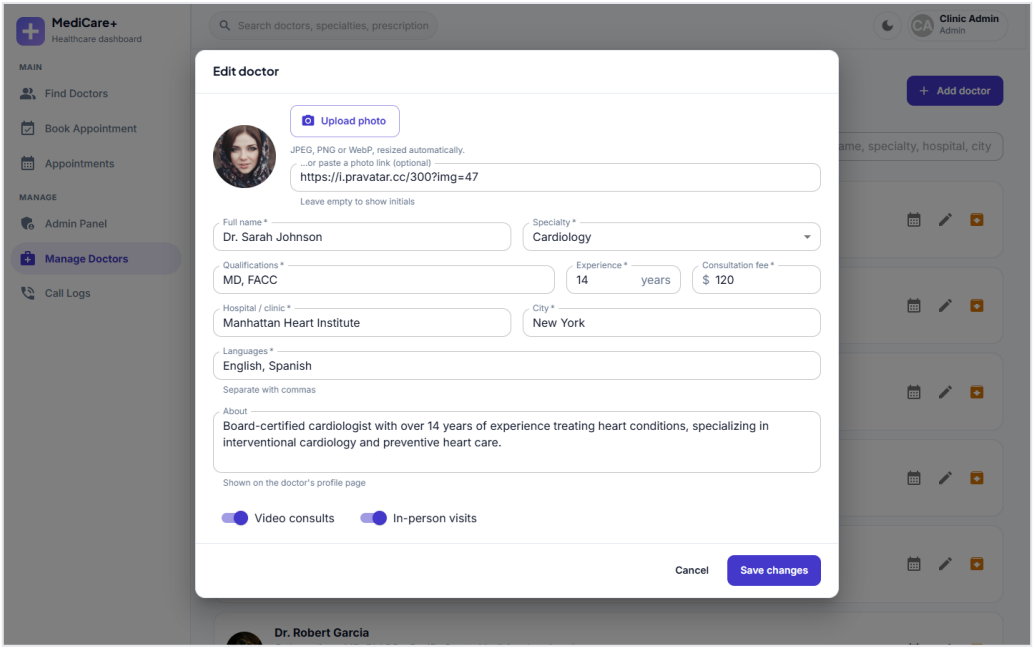


Figure 11 — Add / edit doctor form with photo upload

4.4 Schedules and time off

Click the calendar icon on a doctor. For each weekday, switch work on or off, set start and end time, and optionally a break. Each day shows a live preview such as “15 slots · 09:00 AM–05:00 PM”. Visits are 30 minutes and can be booked up to 7 days ahead.

- **Clinic default** — doctors without their own hours work Monday–Saturday 09:00–17:30 with lunch 12:00–13:00.
- **Time off** — add a holiday, leave or conference by date (with an optional reason). Patients see those days as Closed.
- **Safe changes** — changing hours or adding time off never cancels a booking. The admin is told, for example, “1 upcoming visit is outside the new availability — contact that patient if needed”.

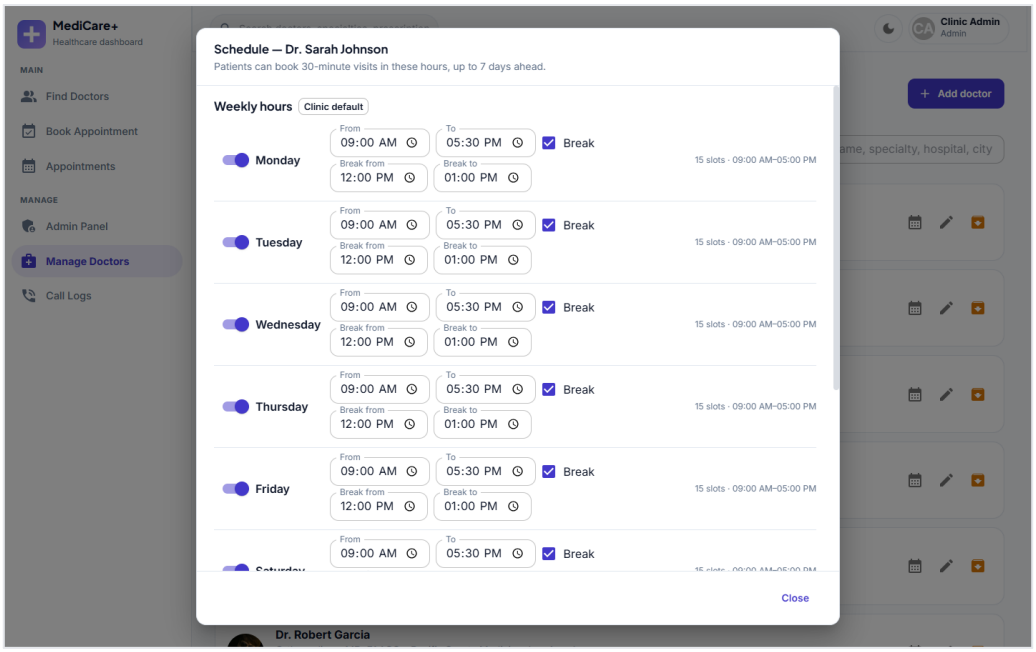


Figure 12 — Weekly hours editor

4.5 Call logs (Telentir)

Shows phone calls handled by the AI voice agent: who called, when, how long, the result, a short summary and the full written transcript. Calls are fetched from Telentir automatically every 60 seconds, or with **Sync from Telentir**. Without a Telentir account the page shows sample calls.

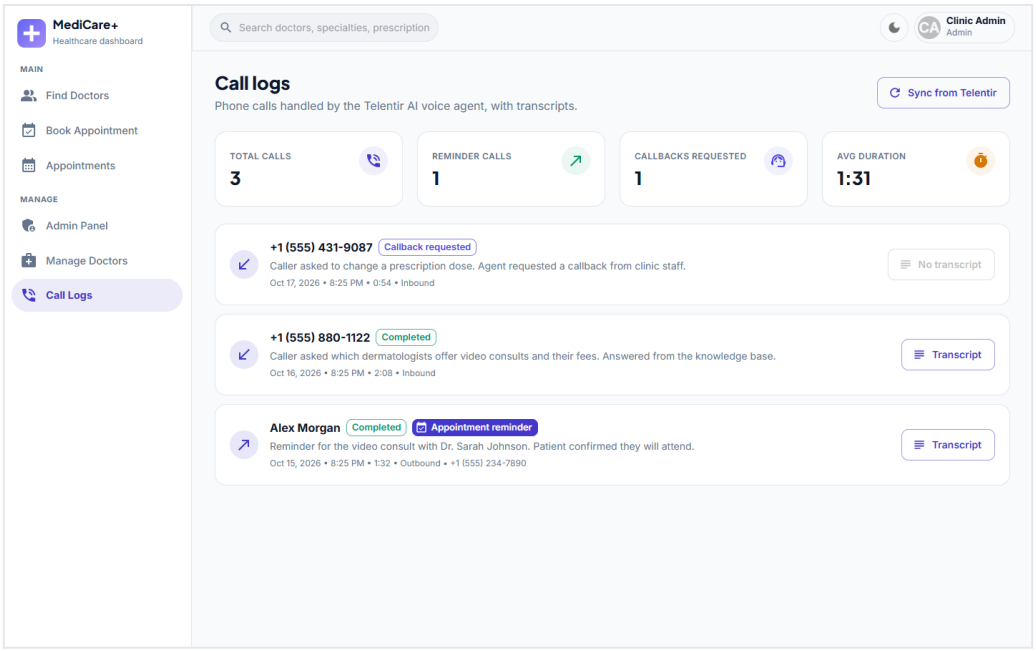


Figure 13 — Call logs

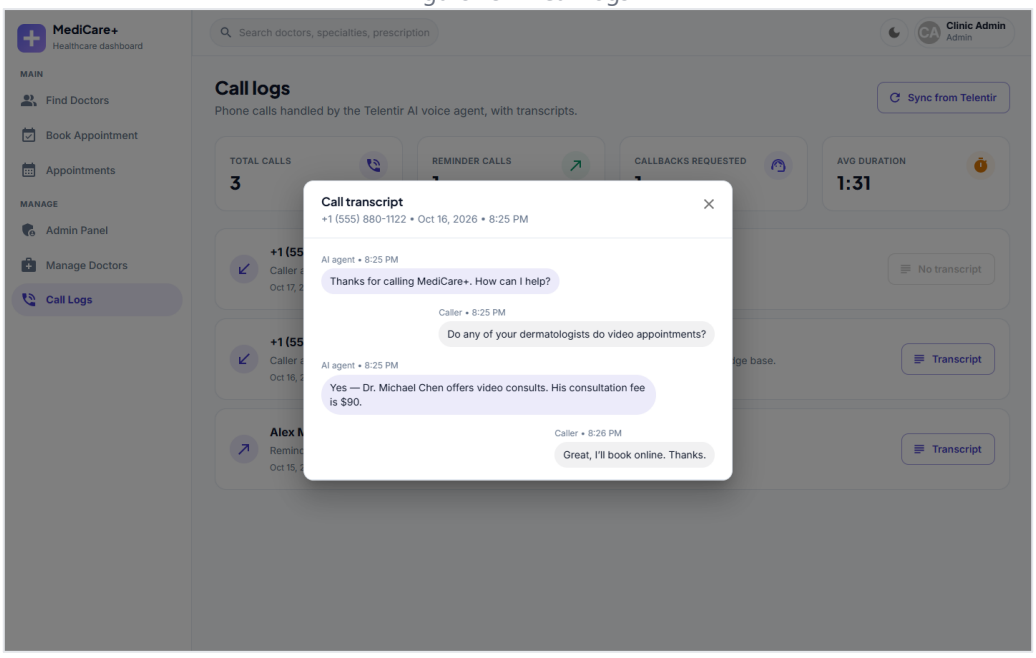
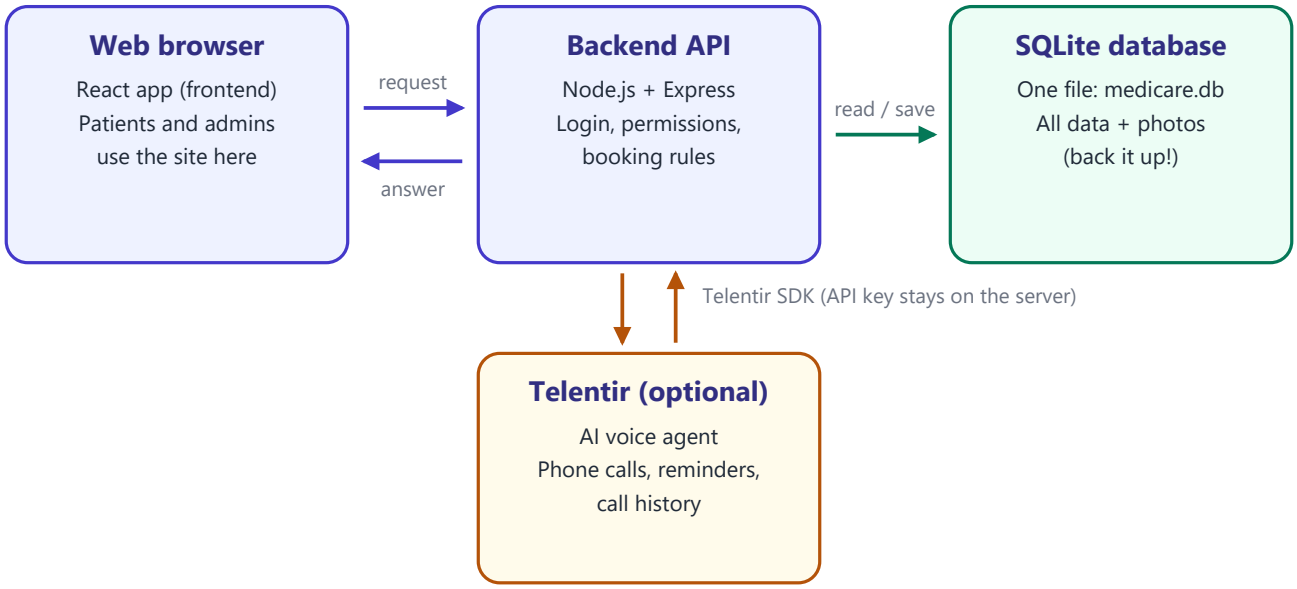


Figure 14 — A call transcript

5. How it works

5.1 The big picture



The **browser** shows the screens. Every time it needs data or wants to change something (for example, book a visit), it sends a request to the **backend API**. The backend checks who is signed in, checks the rules, reads or writes the **database**, and sends back the answer. The browser never talks to the database or to Telentir directly.

5.2 Sign-in and security

- **Passwords** are never stored as written. They are “hashed” with scrypt, a slow one-way method designed for passwords.
- **Sessions** — after signing in, the browser receives a signed token (JWT), valid for 8 hours. Every request carries it; the server checks the signature.
- **Roles** — every request is checked on the server: a patient can only see and change their own records, admins can see everything. Asking for another patient’s record returns “not found”, so record numbers cannot be guessed.
- **Brute-force protection** — 5 wrong passwords for the same email from the same address locks it for 15 minutes.
- **Production safety** — in production mode the server refuses to start without a secret key and admin password, and creates no test accounts.

5.3 Booking rules

- There is **one booking path** for everyone, so the rules can’t be skipped.
- Before booking, the server checks: the doctor exists and is active, offers that visit type, and the slot is open.
- The database itself has a rule that one doctor can’t have two active bookings at the same day and time. So even if two people click at exactly the same moment, only one booking succeeds; the

other sees “That time slot was just taken”.

- Cancelling a visit frees the slot immediately.
- After a booking, the patient gets a notification.

5.4 How open times are calculated

Open times are never typed in by hand. For each of the next 7 days the server takes the doctor’s weekly hours (or the clinic default), splits them into 30-minute slots around the break, and removes: days with time off, slots that are already booked, and times that have already passed today. The same rules file (src/utils/schedule.ts) is used by the server and by the admin’s schedule preview, so the preview always matches reality.

5.5 The database

All data lives in one SQLite file, **server/data/medicare.db**. It is created automatically on first start. Changes to its structure (“migrations”) run automatically and in order.

Table	Holds
users	Accounts: email, name, role (patient/admin), hashed password
patients	Patient details: name, phone, email, optional health details
doctors	Doctor profiles, active/archived status, own weekly hours
doctor_photos	Uploaded doctor photos (stored inside the database)
doctor_time_off	Holidays and leave per doctor
appointments	Every booking with day, time, type, status, reason and fee
prescriptions, medical_records	Patient prescriptions and history
notifications	Messages for each patient
call_logs, reminder_calls	Telentir call history and the reminders we started

> **Back up the database**
> Because everything — including photos — is in this one file, backing up **medicare.db** backs up the whole clinic. It is deliberately not stored on GitHub, because it contains patient data.

5.6 Doctor photos

Photos are shrunk in the browser before upload (max 512 pixels). The server accepts only real JPEG, PNG or WebP files up to 2 MB: it checks the file’s actual content, not its name, so a disguised file is refused. SVG files are not accepted because they can contain scripts.

5.7 Telentir AI voice agent

- **Reminder calls** — when an admin clicks the phone icon, the server asks Telentir's AI agent to call the patient with the appointment details (and never to give medical advice).
- **Call history** — the server collects calls and transcripts from Telentir every 60 seconds and saves them in the database.
- **Knowledge base** — a command uploads the clinic's doctor list to Telentir, so the agent can answer callers' questions about doctors, fees and visit types.
- **Security** — the Telentir API key stays on the server; the browser never sees it.

> **Limit of the current Telentir API**
>
> Telentir's documentation does not describe a way for the AI agent to call this system during a phone call. So callers cannot book directly by phone yet; the agent answers questions and asks staff to call back. This is an open question for Telentir.

6. Technology

Part	Technology	Why
Website	React 18, TypeScript, Vite	Fast, widely used, type-safe
Design	Material UI 6, Framer Motion	Consistent, accessible components; smooth animations
Data loading	TanStack Query, Axios	Caching, automatic refresh, loading states
State	Zustand	Remembers sign-in and theme
Charts	Recharts	Admin dashboard charts
Server	Node.js, Express, TypeScript (tsx)	Simple, reliable REST API
Database	SQLite (better-sqlite3)	One file, no setup, fast
Security	scrypt (Node), JWT (jose)	Safe password storage and signed sessions
AI phone	Telentir SDK (@silyze/telentir-sdk)	Official SDK for the voice agent

7. Installing and running

Needed: **Node.js 20.12 or newer** (22 recommended) and npm. Then, in the project folder:

```
# 1. Install the website and the server
npm install
npm run server:install

# 2. Tell the website to use the server
copy .env.example .env.local        (then set VITE_USE_MOCK=false)

# 3. Start both (two terminal windows)
npm run server      ->  server  at http://localhost:4000/v1
npm run dev         ->  website at http://localhost:5173
```

Open http://localhost:5173 and sign in with a development account:

Email	Password	Role
alex.morgan@example.com	patient-dev-password	Patient (with sample visits and prescriptions)
admin@medicare.example	admin-dev-password	Admin

These test accounts and 10 sample doctors exist only in development. To start with an empty database, stop the server and delete the **server/data** folder.

8. Settings

The server reads its settings from **server/.env** (copy server/.env.example). All are optional in development.

Setting	Needed live	Meaning
PORT		Server port (default 4000)
PUBLIC_URL	Yes	The server's public web address, used in photo links
CORS_ORIGIN	Yes	Web address of the website
DATABASE_PATH		Where the database file is (default server/data/medicare.db)
JWT_SECRET	Yes	Long random secret that signs sign-in sessions
JWT_EXPIRES_IN		How long a sign-in lasts (default 8h)
ADMIN_EMAIL / ADMIN_PASSWORD	Yes	The first admin account (created once)
TELENTIR_API_KEY, TELENTIR_AGENT_ID		Telentir connection (optional)
TELENTIR_KNOWLEDGE_BASE_ID		Knowledge base used by the AI agent (optional)
TELENTIR_SYNC_INTERVAL_SEC		How often to fetch calls (default 60)

The website reads **.env.local**: **VITE_API_URL** (server address) and **VITE_USE_MOCK** (true = built-in sample data with no server, false = use the server).

9. Before going live

- Run the server with **NODE_ENV=production**. It will refuse to start without JWT_SECRET and ADMIN_PASSWORD, and will create no test accounts or sample doctors.
- Serve the website and the server only over **HTTPS**.
- Set **CORS_ORIGIN** to the website's address and **PUBLIC_URL** to the server's address.
- Back up **medicare.db** automatically every day, and test restoring a backup.
- Add the clinic's real doctors in Manage Doctors.
- For real patients in the USA: confirm HIPAA requirements and get a Business Associate Agreement (BAA) from Telentir before turning it on.
- Hide the unfinished buttons listed in section 12 if patients should not see them.

Build the website with **npm run build** (output in the dist folder, served by any web server), and start the server with **npm --prefix server start**.

10. Project structure

```
src/                      Website (React)
  api/                    Talks to the server (one file per area)
  hooks/                  Data loading per area
  pages/                  One folder per page (Login, Dashboard,
                          BookAppointment, ManageDoctors, CallLogs, ...)
  routes/                 Page list + sign-in and role guards
  components/, layouts/   Reusable parts, sidebar, top bar
  types/                  Shared data types (also used by the server)
  utils/schedule.ts       Slot rules shared with the server

server/                   Server (Express + SQLite)
  src/routes/             Every API address and who may use it
  src/auth/               Sign-in, sign-up, passwords, tokens, roles
  src/db/                 Database connection, migrations, sample data
  src/repos/              All database queries, one file per area
  src/booking.ts          The single booking path
  src/telentir/           Call sync, reminder calls, knowledge base
  data/                   The database file (not on GitHub)
```

11. API reference

All addresses start with **/v1**. “Signed in” means any signed-in user; patients only ever receive their own records.

Method	Address	Who
POST	/auth/register, /auth/login	Anyone
GET	/auth/me	Signed in
GET	/doctors, /doctors/:id, /doctors/:id/availability, /doctors/:id/photo	Anyone
GET	/appointments, /appointments/:id	Signed in
POST	/appointments, /appointments/:id/cancel	Signed in
GET	/patients/me, /patients/me/history	Signed in
GET	/prescriptions, /prescriptions/:id	Signed in
GET / POST	/notifications, /notifications/:id/read, /notifications/read-all	Signed in
GET	/admin/stats	Admin
GET / POST / PUT	/admin/doctors, /admin/doctors/:id	Admin
PATCH	/admin/doctors/:id/status (archive / restore)	Admin
PUT / DELETE	/admin/doctors/:id/photo	Admin
GET / PUT	/admin/doctors/:id/schedule	Admin

Method	Address	Who
POST / DELETE	/admin/doctors/:id/time-off, /admin/doctors/:id/time-off/:id	Admin
GET / POST	/calls, /calls/sync	Admin
POST	/appointments/:id/reminder-call	Admin

12. Current limitations

These parts are not finished. They are listed so that nobody mistakes them for working features.

Item	Status
Video consultation	Screen design only — no real video or audio call happens
“Request refill” and “Export” on prescriptions	Buttons do nothing yet
Search bar at the top of every page	Does nothing yet (Find Doctors search works)
“Settings” in the user menu	Does nothing yet
Editing your own profile	Profile is read-only
Email / SMS confirmations	Not sent — confirmations appear in Notifications
Doctor reviews and ratings	No review system; new doctors show “New”
Visit length and booking window	Fixed at 30 minutes and 7 days ahead; one break per day
Booking by phone through the AI agent	Not possible with Telentir’s documented API
Live Telentir calls	Built, but not yet tested with a real Telentir account

13. Suggested next steps

1. **Go live** — host the website and server with HTTPS and set up automatic daily database backups.
2. **Connect Telentir** — add the API key and agent, test a real reminder call, and confirm phone booking support and HIPAA/BAA with Telentir.
3. **Finish the placeholder features** — prescription refill requests, profile editing, the top search bar.
4. **Email or SMS confirmations** for bookings and cancellations.
5. **More schedule options** — different visit lengths, more than one break, booking further ahead.
6. **Real video consultations** using a video-call service.

> **More information**
>
> Source code and full technical README:
> github.com/Leosabinghimire/Doctor-Appointment-Booking-System